

# Serial Protocol Decode — user manual

Ian Ameline · version 1.04 — **beta** · MIT licence

This app turns a Keithley DMM6500 into a serial decoder. Clip onto a UART line, press **Capture**, and read the bytes on the front panel. You do not have to tell it the baud rate, the frame format or which way up the signal is — it works all of that out from the signal itself.

Three things make it worth reaching for over a scope's decode: the **window is large** (a screenful is ~240 bytes, and a recording holds 32 768), every capture is **appended to the USB key automatically** while a key is in the slot, so you have a log without asking for one, and the **serial parameters are detected rather than configured**. For most debugging that is the whole job — connect, press, read.

**This is a beta.** Every number in this manual was measured on a real DMM6500 against a real signal generator unless the text says otherwise — and where something is calculated rather than measured, or untested, it says so in that spot. What that cannot cover is the variety of real devices, so if the app tells you something you can prove is wrong, that is the bug worth reporting. It is built so that **refusing with a reason is acceptable and confident nonsense is not**, and anything it is unsure about it says on the note row at the bottom of the screen.

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## Hooking it up — read this part

**The line must stay between -10 V and +10 V.** The app fixes the meter on its 10 V range. 3.3 V CMOS and 5 V TTL are both fine as they are. For a 12 V line, put a 2:1 resistor divider in front of it — and if it is an open-drain bus, pick resistors big enough not to load it down.

**Connect INPUT LO to the ground of the circuit you are probing.** Everything is measured at INPUT HI relative to INPUT LO, so without a shared ground the readings mean nothing.

**Do not put a differential bus straight across HI and LO.** RS-485, CAN and LIN need to go through a transceiver first, or you need to tap one side of the pair against ground.

**A DC offset does not matter.** The app measures the two voltage levels on your line and puts its decision threshold between them, so it does not care where they sit — only that both fit inside  $\pm 10$  V. A 1.6 V swing sitting up at 6 V decodes fine. **1.6 V is the smallest swing tested.** A 60 mV swing is refused outright.

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## Quick start

1. Connect the line to **INPUT HI** and its ground to **INPUT LO**.
2. Press **Capture** while the device is talking.
3. Read the bytes. Press **View** to switch between hex and plain text.

That is the whole normal workflow. The first capture also locks the baud rate when it is confident, so later captures are faster.

Two more things worth knowing before you need them: **Mode** turns Capture into a recording of up to 32 768 bytes straight to the USB key, still one press — and once a long job starts, **nothing stops it early**, so decide the size before you press — see **The TRIGGER key** below.

## The screen

SERIAL DECODE - 240B FRAMEEnd App

| BAUD  | FORMAT | IDLE | LOGIC    | THRESH | SA/BIT | RATE     | BYTES | ERR | FIT  |
|-------|--------|------|----------|--------|--------|----------|-------|-----|------|
| 19200 | 7E1    | HIGH | 3V3 CMOS | 1.63 V | 8.3    | 160 kS/s | 235   | 0   | 1.00 |

Start bit

|      |                         |                         |                                                   |
|------|-------------------------|-------------------------|---------------------------------------------------|
| 0000 | 20 64 6F 67 2E 20 54 48 | 45 20 51 55 49 43 4B 20 | dog. THE QUICK                                    |
| 0016 | 42 52 4F 57 4E 20 46 4F | 58 20 4A 55 4D 50 53 20 | BROWN FOX JUMPS                                   |
| 0032 | 4F 56 45 52 20 54 48 45 | 20 4C 41 5A 59 20 44 4F | OVER THE LAZY DO                                  |
| 0048 | 47 2E 20 30 31 32 33 34 | 35 36 37 38 39 20 21 22 | G. 0123456789 !"#\$%&'()*+,-./:;<=>?@[\]^_`{ }~th |
| 0064 | 23 24 25 26 27 28 29 2A | 2B 2C 2D 2E 2F 3A 3B 3C | e quick brown fo                                  |
| 0080 | 3D 3E 3F 40 5B 5C 5D 5E | 5F 60 7B 7C 7D 7E 74 68 | x jumps over the                                  |
| 0096 | 65 20 71 75 69 63 6B 20 | 62 72 6F 77 6E 20 66 6F | lazy dog. THE Q                                   |
| 0112 | 78 20 6A 75 6D 70 73 20 | 6F 76 65 72 20 74 68 65 | UICK BROWN FOX J                                  |
| 0128 | 20 6C 61 7A 79 20 64 6F | 67 2E 20 54 48 45 20 51 | UMPS OVER THE LA                                  |
| 0144 | 55 49 43 4B 20 42 52 4F | 57 4E 20 46 4F 58 20 4A | ZY DOG. 01234567                                  |
| 0160 | 55 4D 50 53 20 4F 56 45 | 52 20 54 48 45 20 4C 41 | 89 !"#\$%&'()*+,-./:;<=>?@[\]^_`{ }~the quic      |
| 0176 | 5A 59 20 44 4F 47 2E 20 | 30 31 32 33 34 35 36 37 |                                                   |
| 0192 | 38 39 20 21 22 23 24 25 | 26 27 28 29 2A 2B 2C 2D |                                                   |
| 0208 | 2E 2F 3A 3B 3C 3D 3E 3F | 40 5B 5C 5D 5E 5F 60 7B |                                                   |
| 0224 | 7C 7D 7E 74 68 65 20 71 | 75 69 63                |                                                   |

FRAME HEX pg 1/1 bytes 1-235/235 win 240 [done] log: bytes324 472 B

Capture

View

Mode

NewLog

Save

Options

Figure 1: Main screen, hex view — a 240-byte frame capture, 235 bytes decoded with no errors  
Along the top are the things the app worked out about your line:

| Field              | Meaning                                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------------------|
| <b>BAUD</b>        | The rate it found. A padlock means it is locked and will not be re-detected.                                |
| <b>FORMAT</b>      | Data bits, parity, stop bits — 8N1, 7E1 and so on.                                                          |
| <b>IDLE</b>        | What the line rests at: HIGH for CMOS/TTL, LOW for an RS-232-style line.                                    |
| <b>LOGIC</b>       | The family it looks like, e.g. 3V3 CMOS, 5V TTL.                                                            |
| <b>THRESH</b>      | The voltage it is using to tell 0 from 1.                                                                   |
| <b>SA/BIT</b>      | Samples per bit. More is better; 8 is comfortable, 4 still works.                                           |
| <b>RATE</b>        | How fast the meter sampled.                                                                                 |
| <b>BYTES / ERR</b> | How many bytes came out, and how many of them you cannot trust.                                             |
| <b>FIT</b>         | How well one bit-length explains what it measured, from 0 to 1. Below about 0.9 means the timing was messy. |

A dash means “not measured yet”.

**Broken bytes are shown, not hidden.** ERR turns red when there are any, and every row containing one is drawn red across its whole width — offset, hex and ASCII together — so damage is found by looking rather than by counting. Flagged bytes read as ? in the ASCII gutter:

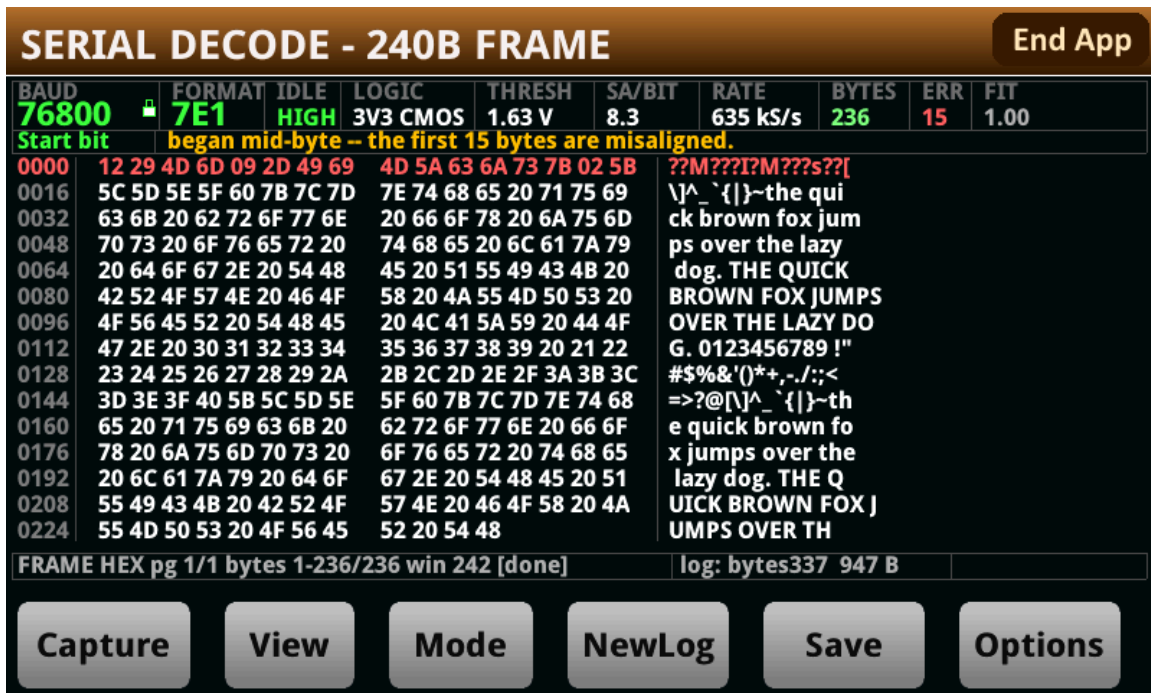


Figure 2: A capture that began mid-byte: the note names 15 bytes and ERR reads 15

That is a real 240B capture that began part way through a byte. **The note row names the cause** — began mid-byte -- the first N bytes are misaligned. — which is the difference between a puzzle and a known condition: the app arms on a busy line, so it lands wherever the traffic happens to be, and the bytes before the first gap between messages cannot be framed. On a busy line roughly one capture in eight starts this way.

**The note and ERR agree, and that is deliberate.** The note's number is how far the damage reaches: the last byte in the opening region the decoder could not frame, so "the first N" is literally true and nothing past N is being accused. ERR is then never less than N. Above, the note says 15 and ERR says 15 — while only 11 of those bytes actually failed a check. The other four are the reason ERR counts the whole region: inside a misaligned head the bit boundaries are in the wrong places, so a frame there can satisfy its parity and stop bit by luck and still hand back a plausible wrong byte. 3, 7, 9 and 13 came out as M, I, M and s — perfectly ordinary characters, and wrong. **ERR counts bytes you cannot trust, not checks that failed**, because the silently wrong ones are the dangerous ones.

Two exclusions remain, and both keep ERR at 0 on good data. When no misaligned head is identified, the first three frames are ignored — a gapless line has no idle for the framer to anchor on, so it routinely resynchronises a frame or two in, and counting that made ERR read 1 or 2 on every healthy capture. The final frame is ignored always, because the capture boundary cuts it in half. Neither exclusion applies across a head the app has evidence for, which is what stopped the panel reading ERR 1 beside a note naming four bytes.

Everything from the first gap onward decoded cleanly, and the log on the USB key holds every byte: one that failed its parity or stop bit is still written, marked, rather than dropped.

Underneath: the **trigger cell**, the bytes, then two rows of text. The **status row** tells you where you are (FRAME HEX pg 1/1 bytes 1-155/155 win 240 [done]) and names the log file.

**Which mode you are in is in the title bar** — SERIAL DECODE - 240B FRAME, - 8K CAPTURE or - 32K CAPTURE, in the largest type on the screen.

**The cell at the left of the note row is what a capture WAITS for**, colour-coded so a glance is enough: green Start bit (the normal case — it waits for the line's own start bit), yellow Free run (it does not wait for anything), cyan Trigger key (it waits for the physical key).

**The cell at the bottom right is the rear BNC**, and it is **empty when the rear BNC is off** — which is the default. When it is not, it names the direction: green EXT TRIG IN, amber EXT TRIG OUT, cyan EXT TRIG I/O. A trailing ? means the input is switched on but *not in force*, which happens under Free run — free run means do not wait, so there is nothing for the input to gate.

During a recording the status row carries a live counter — recording... 40 % of the buffer, then decoding... 4096/8192 bytes — with a **cyan progress bar** beside it, and the cell to its right reads Capture or Mode = Stop.

**View** switches the byte display to plain text, which is what you want for anything human-readable — and a row carrying a flagged byte is red here too:



Figure 3: Main screen, text view — the first five bytes misaligned, ERR agreeing at 5

**The note row is the important one.** Warnings appear there — an ambiguous baud rate, a line that disagrees with the rate you locked, a recording that stopped early. If more than one applies it ends with (+2 more), and pressing **Save** writes all of them to a file.

Here it is doing that — a capture that started part way through a byte, which the app detected and said so rather than showing two wrong bytes without comment:

**SERIAL DECODE - 240B FRAME**
End App

| BAUD                                                                 | FORMAT                  | IDLE                    | LOGIC                                                                                                                                                                                                             | THRESH | SA/BIT | RATE     | BYTES | ERR | FIT  |
|----------------------------------------------------------------------|-------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|----------|-------|-----|------|
| 76800                                                                | 7E1                     | HIGH                    | 3V3 CMOS                                                                                                                                                                                                          | 1.63 V | 8.3    | 635 kS/s | 236   | 2   | 1.00 |
| <b>Start bit</b> began mid-byte -- the first 2 bytes are misaligned. |                         |                         |                                                                                                                                                                                                                   |        |        |          |       |     |      |
| 0000                                                                 | 07 53 55 4D 50 53 20 4F | 56 45 52 20 54 48 45 20 | ??UMPS OVER THE                                                                                                                                                                                                   |        |        |          |       |     |      |
| 0016                                                                 | 4C 41 5A 59 20 44 4F 47 | 2E 20 30 31 32 33 34 35 | LAZY DOG. 012345                                                                                                                                                                                                  |        |        |          |       |     |      |
| 0032                                                                 | 36 37 38 39 20 21 22 23 | 24 25 26 27 28 29 2A 2B | 6789 !"#\$%&'()*+,-./:;<=>?@[\\]^_`{ }~the quick brown fox jumps over the lazy dog . THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG. 0123456789 !"#\$%&'()*+,-./:;<=>?@[\\]^_`{ }~the quick brown fox jumps over the |        |        |          |       |     |      |
| 0048                                                                 | 2C 2D 2E 2F 3A 3B 3C 3D | 3E 3F 40 5B 5C 5D 5E 5F |                                                                                                                                                                                                                   |        |        |          |       |     |      |
| 0064                                                                 | 60 7B 7C 7D 7E 74 68 65 | 20 71 75 69 63 6B 20 62 |                                                                                                                                                                                                                   |        |        |          |       |     |      |
| 0080                                                                 | 72 6F 77 6E 20 66 6F 78 | 20 6A 75 6D 70 73 20 6F |                                                                                                                                                                                                                   |        |        |          |       |     |      |
| 0096                                                                 | 76 65 72 20 74 68 65 20 | 6C 61 7A 79 20 64 6F 67 |                                                                                                                                                                                                                   |        |        |          |       |     |      |
| 0112                                                                 | 2E 20 54 48 45 20 51 55 | 49 43 4B 20 42 52 4F 57 |                                                                                                                                                                                                                   |        |        |          |       |     |      |
| 0128                                                                 | 4E 20 46 4F 58 20 4A 55 | 4D 50 53 20 4F 56 45 52 |                                                                                                                                                                                                                   |        |        |          |       |     |      |
| 0144                                                                 | 20 54 48 45 20 4C 41 5A | 59 20 44 4F 47 2E 20 30 |                                                                                                                                                                                                                   |        |        |          |       |     |      |
| 0160                                                                 | 31 32 33 34 35 36 37 38 | 39 20 21 22 23 24 25 26 |                                                                                                                                                                                                                   |        |        |          |       |     |      |
| 0176                                                                 | 27 28 29 2A 2B 2C 2D 2E | 2F 3A 3B 3C 3D 3E 3F 40 |                                                                                                                                                                                                                   |        |        |          |       |     |      |
| 0192                                                                 | 5B 5C 5D 5E 5F 60 7B 7C | 7D 7E 74 68 65 20 71 75 |                                                                                                                                                                                                                   |        |        |          |       |     |      |
| 0208                                                                 | 69 63 6B 20 62 72 6F 77 | 6E 20 66 6F 78 20 6A 75 |                                                                                                                                                                                                                   |        |        |          |       |     |      |
| 0224                                                                 | 6D 70 73 20 6F 76 65 72 | 20 74 68 65             |                                                                                                                                                                                                                   |        |        |          |       |     |      |

FRAME HEX pg 1/1 bytes 1-236/236 win 242 [done]
 

log: bytes337 2843 B

Capture
View
Mode
NewLog
Save
Options

Figure 4: The note row reporting a mid-byte start

**An empty note row is not a promise that everything is perfect.** It means the app did not detect anything wrong. A noise spike landing inside a data bit changes that byte with nothing to give it away.

## The buttons

| Button           | Does                                                                                                                                           |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Capture</b>   | Take a capture and decode it. In a recording mode, does the whole recording.                                                                   |
| <b>View</b>      | Switch the display: hex + ASCII → plain text → back.                                                                                           |
| <b>Mode</b>      | Choose what Capture does: a screenful, an 8 kB recording, or a 32 kB one.                                                                      |
| <b>NewLog</b>    | Start a new numbered log file on the USB key.                                                                                                  |
| <b>Save</b>      | Write a full report of what is on screen to the USB key.                                                                                       |
| <b>Options</b>   | The settings screen.                                                                                                                           |
| <b>Up / Dn</b>   | Right-hand edge. Only appear when there is more than one page. Once you press either, <b>page N of M</b> appears at the right of the note row. |
| <b>Lock Rate</b> | Right-hand edge. Only appears when the rate is not locked.                                                                                     |

Every button responds immediately except the ones that take a capture. **Capture takes about 1.7 seconds when the rate is locked, and 4 to 9 seconds when it is not** — because to find an unknown rate it has to capture two or three times. Slower lines take longer, not less. The exact time varies from press to press because it depends on where the traffic happens to be when you press.

**A press during a capture is not lost — it is queued** and acted on when the capture finishes. So if the panel seems to ignore you, wait rather than pressing again.

## Paging through a long capture



Figure 5: Page 4 of 35 of a finished 32 kB recording, hex view

A 32 kB recording is far more than one screen, so **Up** and **Dn** appear down the right-hand edge and **page N of M** appears at the right of the note row — the two numbers in white, the words dimmed, so the count reads at a glance. The note beside it says which bytes are on screen and that the whole run is in the file, which is the thing you cannot otherwise tell from a screenful taken out of the middle:

panel shows bytes 24577-32768 of 32768; all are in bytes340.txt

In the figure that note is abbreviated, ending (+1 more): this run also began mid-byte, and two notes do not fit one row. **Save** writes both out in full.

The offset column agrees with it — the rows start at 25296, not at zero — and the status row carries the run's own summary: 32768 bytes, 6 it cannot vouch for, 68 windows, and the file it went to. **That 6 is the same figure as ERR**, not a second opinion about it.

**The page count follows the view.** Press **View** on that same capture and it becomes 7 pages instead of 35, because a text row holds 80 characters and a hex row holds 16 bytes — 1200 bytes a page against 240:

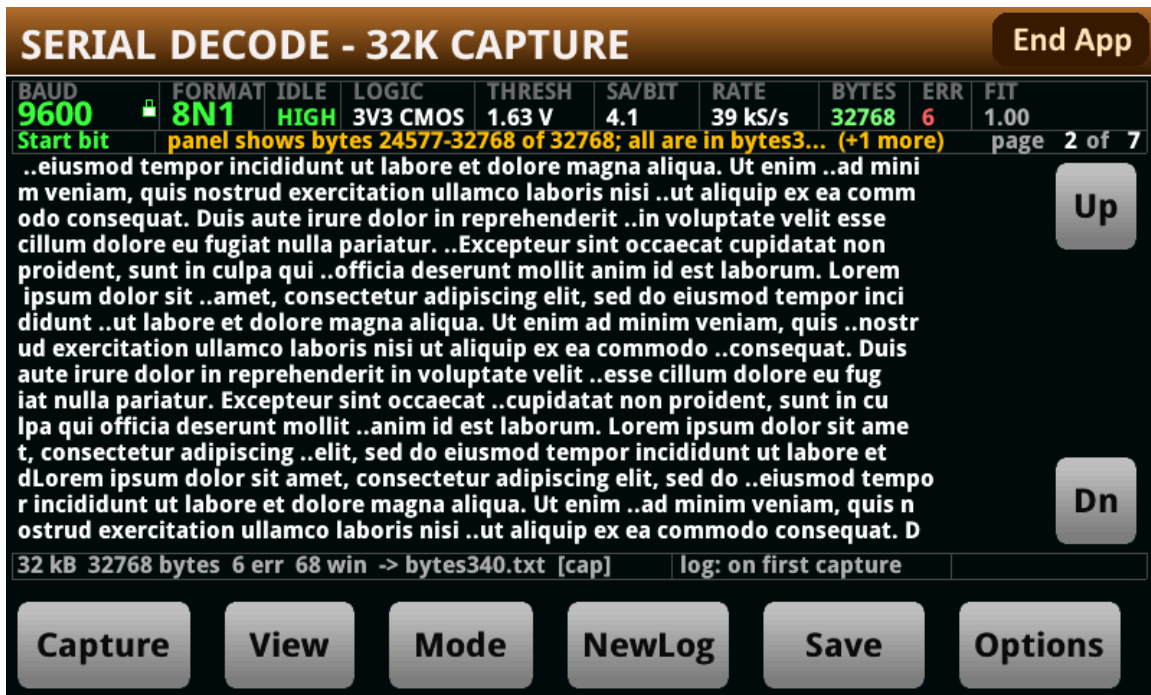


Figure 6: The same 32 kB capture in text view: page 2 of 7

Nothing about the capture changed; only how much of it fits on a screen. The byte range in the note row is the same, and so is the summary.

## The TRIGGER key — the one hardware button that matters

The **TRIGGER** key is the physical key on the front panel, to the **right** of the screen — the lowest of that column of buttons. It is not part of the app’s own row of buttons. It was intended to do two things nothing on the glass can do; **one of them works** — timing a capture — and the other, stopping a long job, does not. Both are described below, because knowing which is which is the difference between waiting confidently and pressing a key that does nothing.

### 1. It does not stop a long job, and nothing else does either

**A recording runs to its own end.** Once you press **Capture** in 8 kB or 32 kB, nothing on the instrument will cut it short — not a button on the glass, and not the **TRIGGER** key. Decide before you press. The status row says so before you commit:

32 kB ready -- 32768 bytes to a file, no stop once started

**There is deliberately no time estimate.** Two different durations apply and only one is knowable here: the time your *device* takes to send that many bytes, and the meter’s decode, which costs several times more — 8 kB is a couple of seconds on the wire and around half a minute to decode. A single number would be reassuring and wrong, so what you get is the exact byte ceiling and a progress bar.

**What does bound a run,** so it can never hold the panel indefinitely:

- the byte ceiling for the mode — 8192 or 32768;

- a quiet line: one second with no activity ends it;
- under flow control, 32 windows or 20 minutes of wall clock, whichever comes first. The 20 minutes is what matters on slow lines, where a full 32 kB window would take 18 minutes at 300 baud.

**A tap on Capture during a run is not a stop.** It is queued by the instrument and delivered when the run finishes. The app recognises presses made during the run it has just finished and discards them, saying so on the note row, rather than starting another recording you did not ask for.

## 2. It can time a capture

Set Options ► Trigger = Trigger key, press **Capture**, and the capture waits for you: it fires when you press the physical key, so you decide the moment the line is sampled. Verified on this hardware — armed on a deliberately idle line, the capture fired on the key press rather than timing out.

If the trigger you chose never arrives, the capture goes ahead free-running after a few seconds rather than failing, and the note row says which one was missing.

These two jobs never collide: the key times a capture only when you have asked for that in Options, and stops a run only while one is going.

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## How much you can capture at once

Four answers, and almost everyone only ever needs the first. **All of them are one press.**

| You need                          | Use                                      | Get                                        |
|-----------------------------------|------------------------------------------|--------------------------------------------|
| a look at what a device is saying | <b>FRAME</b> — just press Capture        | ~ <b>240 bytes</b> , on screen             |
| a longer burst, quickly           | <b>Mode</b> → 8 kB, then Capture         | up to <b>8 192 bytes</b> , to a file       |
| the longest single recording      | <b>Mode</b> → 32 kB, then Capture        | up to <b>32 768 bytes</b> , to a file      |
| more than that, losslessly        | flow control, and a device that obeys it | as much as you like, ~20 minutes per press |

**FRAME is the normal case and covers almost all debugging.** One press gives you a screenful, already decoded, with the baud rate and frame format worked out for you and a copy appended to the USB key automatically. You do not have to set anything up or lock anything down.

## Recording to a file

Press **Mode** until the cell reads 8 kB or 32 kB. **You must lock the baud rate first** — press **Lock Rate**, or type the rate into **Options**. Then **press Capture once**, and that press does the whole job: it records, decodes every byte, writes them to the USB key, and comes back with the tail on screen. There is nothing to press in the middle.

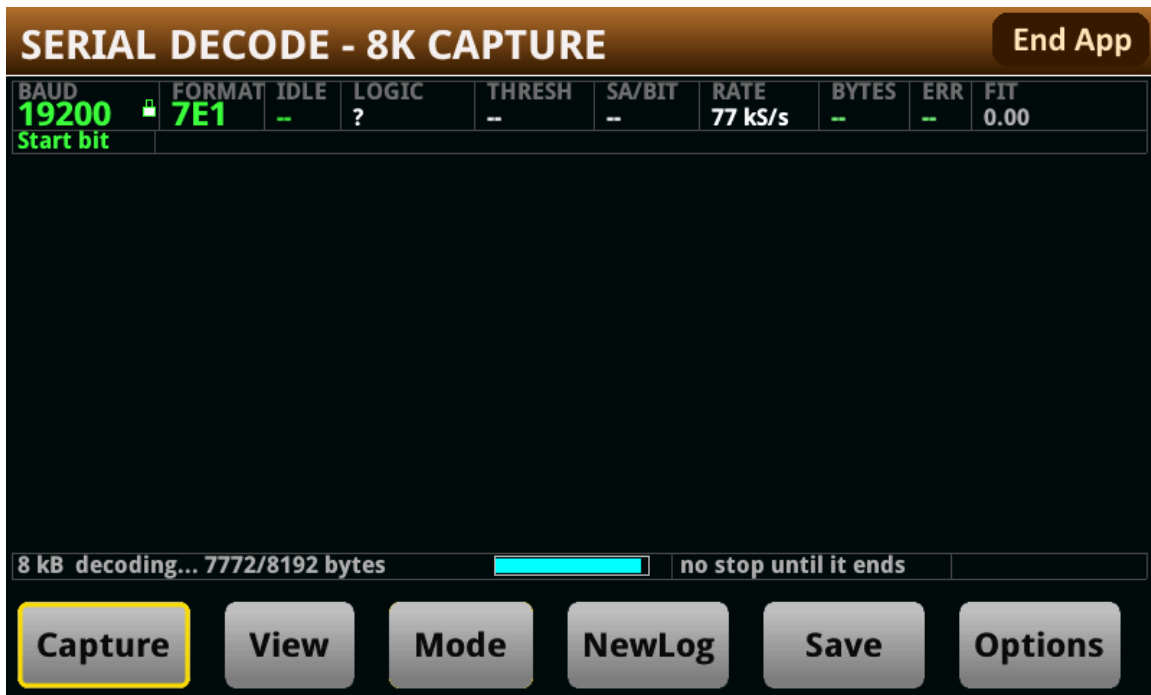


Figure 7: An 8 kB recording, part way through its decode

While it works, the counter on the status row moves — first recording... N % of the buffer, then decoding... N/8192 bytes — with a cyan progress bar beside it, and the cell to its right says whether anything can stop it. The screenshot above is a real 8 kB run caught at 7772 of 8192 bytes decoded; the dump area is blank because nothing has been shown yet, and the note row is empty because there is nothing wrong. **There is no way to finish it early** — see the section on the TRIGGER key above for what bounds it instead.

**When it finishes, the note row goes quiet.** The status row's summary — bytes, errors, window count and filename — is the completion message; the yellow note row is reserved for what went wrong, so an empty one after a recording means there was nothing to say.

It stops by itself when the window is full or when the line has been quiet for a second. There is also a 20-minute backstop, so nothing can wait for ever — it only matters on very slow lines, where a full 32 kB window takes 18 minutes at 300 baud.

**Nothing is dropped inside a recording.** What you get is a continuous, in-order run of the line — verified with a 1024-byte non-repeating test pattern, where the decoded bytes were one unbroken slice of it.

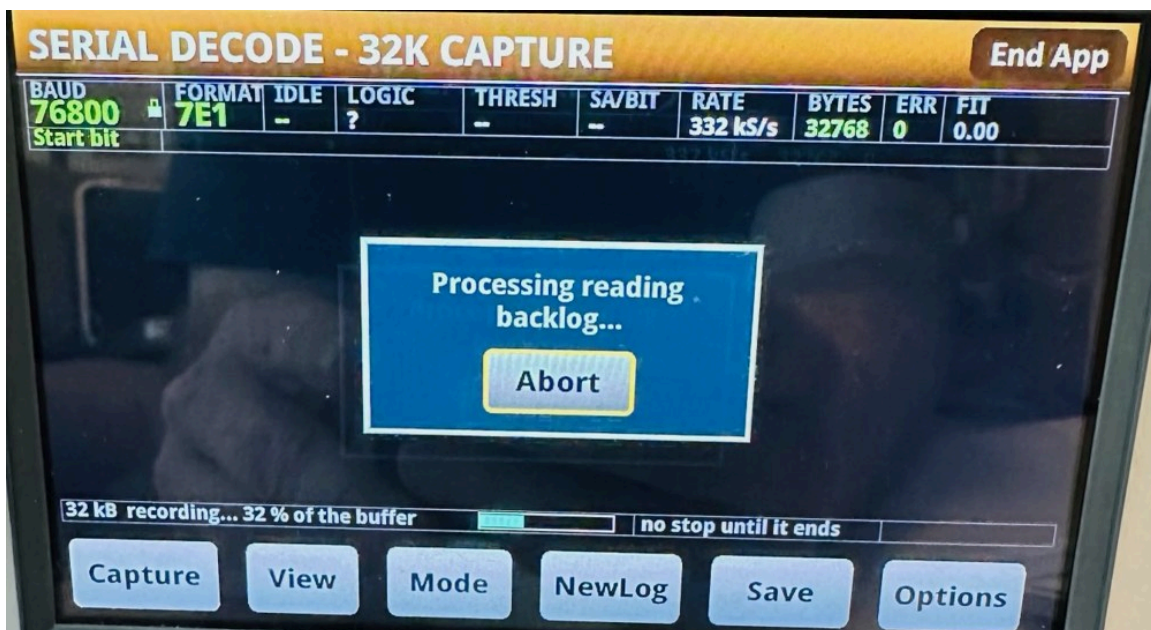


Figure 8: The instrument's own backlog dialog, over a 32 kB recording at 76.8 kBd

**If a Processing reading backlog... dialog appears, ignore it. Do not press Abort.**

It is the instrument's own message, not the app's: the digitizer is producing readings faster than the meter is moving them out of the buffer. It **clears by itself** and the recording carries on normally.

Seen on the bench from 38.4 kBd upwards — at 182 kS/s, at 635 kS/s recording a 76.8 kBd line, and within a second of pressing Capture at 1 MS/s on a 115.2 kBd line. Every one of those recordings completed and decoded correctly, including a full 32 kB one.

Pressing **Abort** would stop the acquisition the dialog is describing, which is the one thing that actually loses the capture. There is nothing to do but let it run.

The photograph above is that dialog over a live 32 kB recording of a 76.8 kBd line at 332 kS/s, with the progress bar behind it at 32 % of the buffer and the row beside it reading no stop until it ends. It is a photograph rather than a screenshot for a reason worth knowing: **the instrument's web interface stops answering while the digitizer runs**, so the remote screen grab cannot capture this dialog at all – eight attempts timed out. If you need to document or report something that happens during an acquisition, a camera is the only way to see it.

### Which window size to pick

The window is the unit of work: one press records a window, decodes it, and files it. So the size is a straight trade, and neither choice affects what gets decoded — both are lossless.

|                         | 8 kB                                        | 32 kB                                   |
|-------------------------|---------------------------------------------|-----------------------------------------|
| Bytes per press         | 8 192                                       | 32 768                                  |
| Recording, at 9600 baud | ~8.5 s                                      | ~34 s                                   |
| Decoding afterwards     | ~30 s                                       | ~2 minutes                              |
| Best for                | seeing something soon, and<br>stopping soon | the most data for the fewest<br>presses |

**Decoding is the slow part**, and it is slower than the line: the meter decodes roughly 300 bytes a second whatever the baud rate. That is why 8 kB feels much more responsive even though both are one press.

**The app starts in FRAME**, and one press of **Mode** reaches 8 kB, a second 32 kB. Stop at 8 kB when you would rather see bytes soon than see all of them, or while you are still working out whether you are probing the right pin; go on to 32 kB when you want the most data for the fewest presses.

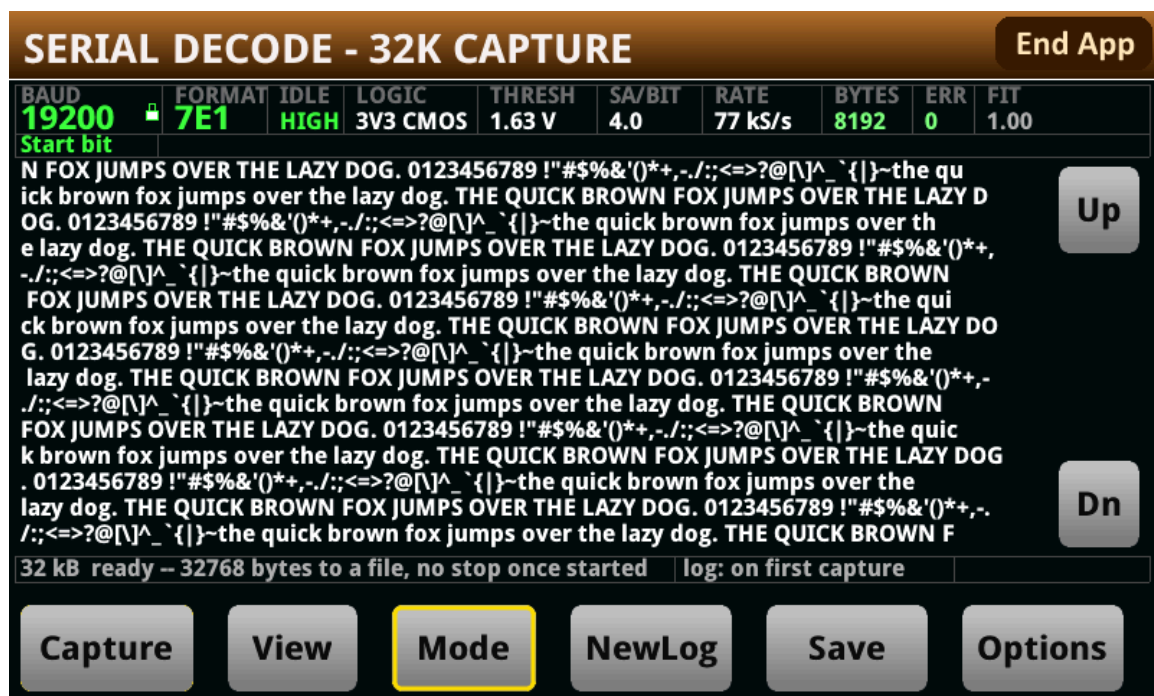


Figure 9: After an 8 kB recording, with Mode pressed once to line up the next one

Above: an 8 kB recording has finished and **Mode** has been pressed once, so the title bar reads 32K CAPTURE while the previous run's retained tail is still on screen. Note what the panel is saying – Up and Dn have appeared because the tail is more than one page, the status row states the next run's ceiling and that nothing will stop it, and **the note row is empty**, because a recording that worked has nothing to warn about.

**The mode you choose stays chosen.** The title bar says which one you are in — SERIAL DECODE - 240B FRAME, - 8K CAPTURE or - 32K CAPTURE — and a recording that finishes leaves it alone, so pressing Capture again records another window. Only **Mode** changes it. Two things move it for you, and both say so on the note line: **Options ▶ Auto Detect**, because auto-detection cannot run in a recording mode at all, and clearing the baud rate, because a recording needs it locked.

### **More than one window: flow control**

A window is not a total. If your device is built to wait, you can capture as much as you like without losing a byte, **and without pressing anything per window** — up to 32 windows or 20 minutes per press, whichever comes first. To carry on past that, press **Capture** again; it resumes where it stopped, in a new file.

Set Options ▶ Rear BNC = FC Out. The rear EXT TRIG OUT then emits **one ~5 V, ~10 µs pulse each time a capture is armed** (measured on a scope: 4.92 V, 9.4–9.8 µs — the width is the meter’s own default and this firmware does not let the app change it) — a credit meaning “*send now, and send no more than one window’s worth*”. Then one press of **Capture** runs the whole conversation: credit, record, decode, file, credit again — round and round, with the window number on the status row, until the device stops sending. While the app is decoding, nothing is armed, no credit is issued, and a device that waits stays quiet, so the gap where bytes could be lost is closed by design. Everything lands in one file, in order.

It ends when your device goes quiet after a credit, or when it reaches one of two backstops: **32 windows** or **20 minutes**. There is no key that ends it early. Those exist so a device that never stops talking cannot leave the panel busy for ever, and the 20 minutes is usually the one you meet — decoding a window takes longer than recording it, so 20 minutes is about eight windows at 9600 baud rather than 32.

If it stops for either reason it says which, and it tells you how to carry on: **press Capture again**. A finished recording leaves you in the mode you chose — the title bar still reads SERIAL DECODE - 32K CAPTURE — so it is one press, and the bytes that follow go into a **new** numbered file rather than the one it was just writing.

**No bytes are lost in the gap.** Your device only transmits when it is credited, and no credit goes out until the next recording arms, so it is still waiting exactly where the run stopped. What you get is one conversation split across two files, in order, with nothing missing between them.

Three things your device has to get right:

- **Catch a 10 µs pulse.** That is short: use an interrupt, an edge-triggered input or a hardware latch. A firmware polling a GPIO in a loop can miss it, and at 250000 baud 10 µs is only 2.5 bit times. It is a pulse per window, not a level you can poll once.
- **Send no more than one window per pulse** — 8 192 or 32 768 bytes depending on the mode. The pulse carries no size information, so the limit has to be built into your firmware.
- **Go quiet when it has nothing to send.** That silence is what tells the app the transmission is over.

**Honest limitation:** the pulse itself is verified — 4.92 V, ~10  $\mu$ s, one per arm, measured on a scope — and the loop that issues them is tested. The full round trip, with a device that actually waits for the pulse, has **not** been tested here, because the signal generator on this bench cannot be gated by the meter’s trigger output. The mechanism is sound by construction; you would be the first to close the loop.

### How long a recording gets you

There are two different times here and it is worth keeping them apart:

- **Signal** — how much of the line ends up in the file.
- **Waiting** — how long you stand there before the buffer is full.

Below about 19200 baud they are the same. Above it they are not, because the meter writes about **100 000 readings a second** into its memory however fast the digitizer is told to sample. The samples are properly spaced and the record is complete; gathering them just takes longer than the stretch of line they represent, so you stand there longer than the signal you get.

That ceiling is a separate thing from the sample rate, and the two are easy to confuse. The **sample rate** decides how many samples land in each bit — a frame capture gets very nearly the rate it asks for, a recording’s burst gets somewhat less, and that difference is what limits recording to 153600 baud (above). The **100 000 readings a second** is not about fidelity at all: it is how fast readings reach memory, and it decides only how long you wait.

| Locked rate | Signal you get | Waiting for the full 32 kB |
|-------------|----------------|----------------------------|
| 300 Bd      | 1092 s         | the same                   |
| 1200 Bd     | 273 s          | the same                   |
| 2400 Bd     | 136 s          | the same                   |
| 4800 Bd     | 68 s           | the same                   |
| 9600 Bd     | 34 s           | the same                   |
| 19200 Bd    | 17 s           | the same                   |
| 38400 Bd    | 8.5 s          | about 17 s                 |
| 57600 Bd    | 5.7 s          | about 17 s                 |
| 115200 Bd   | 2.8 s          | about 18 s                 |
| 153600 Bd   | 2.1 s          | about 21 s                 |

Every row is the same 32 768 bytes — what changes is how long that takes to go past. The 8 kB window is a quarter of each figure. Then add the decoding, which does not depend on the baud rate: about two minutes for 32 kB, half a minute for 8 kB.

**You do have to fill it.** There is no way to stop a recording early — see the TRIGGER key section above — so choose the window with the whole wait in mind.

**153600 baud is the fastest rate that can record. Above that, FRAME only.**

The reason is that a recording and a frame capture do not get the same sample rate out of the same request. A frame capture gets very nearly the 1 MS/s it asks for; a recording's burst acquisition loses about half a microsecond per sample, so the same request delivers about 662 kS/s. The floor is four samples per bit, and the crossover works out at **165563 baud** — so of the standard rates, 153600 is the last one that clears it, at 4.3 samples/bit.

Above it the app **refuses before you press**, on the note row, naming the figure a recording would actually get:

192000 Bd: a recording gets 3.4 samples/bit, under the 4 floor -- use FRAME

That figure is deliberately the recording's, not the header's. The SA/BIT cell will read about 5.2 at 192000 baud, which is true of the frame capture it was measured from — the two are different acquisitions, and the note says which one it means.

**FRAME keeps working to the full 250000 baud**, and usefully so: at 192000 baud a frame capture holds 384 bytes.

---

## Reading the bytes

The hex view shows two groups of eight bytes and then the same bytes as text. The offset down the left is dimmed because it is there to navigate by, not to read. A byte that could not be recovered shows as - -.

**ERR 0 does not mean the bytes are definitely right.** It counts broken stop bits and parity failures. Noise landing inside a data bit flips it with nothing to detect — which is why the note row matters more than the error count.

If you see **began mid-byte — the first N bytes are misaligned.**: on a line that never goes quiet, the capture starts wherever it starts, and if that is halfway through a byte the first few bytes are chopped up wrongly. They are shown rather than hidden, but **do not trust their values**. Everything after the first gap in the traffic is fine. On traffic with normal gaps between messages it costs **5 to 12 bytes**; on a line that never pauses it can be far more — **up to 165 bytes measured** on back-to-back text at 250000 baud, because there is no gap to resynchronise on. A device that pauses between messages never shows it at all.

---

## Locking the baud rate, and the amber padlock

The app locks the rate for you after a good capture, which is why later captures are quicker. It only does this when it is confident: the rate landed on a standard one, enough clean frames came through, FIT was 0.9 or better, and the voltage levels were steady.

When it is not confident it leaves the padlock **amber** and offers you the **Lock Rate** button. The bytes are still shown — they just are not trusted enough to carry forward. Press **Lock Rate** to pin it anyway, or capture again.

**Auto-lock never locks polarity.** It is worked out fresh on every capture, on purpose: guessing it wrong once and then sticking to it is how you get a confident, completely wrong decode.

(**Lock Detected** on the Options screen *does* copy the detected polarity into the fields along with everything else — but there you can see what it picked before you commit it.)

### If the line changes rate, the app gives the lock up

A locked rate is a promise about the wire, and when the wire stops keeping it the lock is worse than useless — it turns every frame into an error and nothing gets better by pressing Capture again. So when **all three** of these hold, the app **drops the lock by itself** and re-detects from the same samples:

- a rate is locked;
- the bit time it measures on the wire disagrees with that rate;
- more than a quarter of the frames in the middle of the capture failed.

It says so on the note row, naming the rate it discarded — 19200 baud fit nothing -- unlocked, and this capture reads 38400 — and from there auto-lock pins the new rate on the usual terms. That is the same result as typing 0 into Options by hand, which is what you had to do before.

**The middle condition is what protects a rate you typed deliberately.** If the rate fits the wire and the *format* is wrong, every frame fails too — but the measurement agrees with your number, so your number is kept.

---

## Options

**SERIAL DECODE OPTIONS** End App

|                                               |            |
|-----------------------------------------------|------------|
| <b>Baud Rate</b><br>0 = auto-detect           | 0.00000Bd  |
| <b>Data Bits</b><br>auto, or fix it           | Auto (7/8) |
| <b>Parity</b><br>auto or fixed                | Auto       |
| <b>Polarity</b><br>idle level                 | Auto       |
| <b>Trigger</b><br>capture start               | Start bit  |
| <b>Rear BNC</b><br>trig in / flow-control out | Off        |

Auto Detect Lock Detected Apply Cancel

Figure 10: Options screen

| Setting          | What it does                                                                                                                                     |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Baud Rate</b> | 0 means auto-detect. Type a number to lock that rate; it is rounded to a whole baud.                                                             |
| <b>Data Bits</b> | Leave on Auto (7/8). Only use Auto (any) if you know the device sends 9-bit words — it can make a damaged capture look tidier while being wrong. |
| <b>Parity</b>    | Auto, or pin None / Even / Odd.                                                                                                                  |
| <b>Polarity</b>  | Auto, Idle high (normal CMOS/TTL) or Idle low (inverted). Auto is reliable unless the line is almost never idle.                                 |

Each setting is independent — pinning one leaves the rest on auto.

**Apply** commits your changes and takes a capture. **Cancel** puts everything back to how it was when you opened the screen. **Auto Detect** forgets everything locked, takes a fresh capture, and locks whatever that finds — it is the quick way back when you have locked onto the wrong thing. **Lock Detected** copies the last capture's settings into the fields so you can look at them before committing.

There is no stop-bit setting. A second stop bit looks exactly like a bit of idle line, which the decoder already copes with, so a 2-stop device decodes correctly and is reported as 1 stop.

### Advanced

Two settings you can ignore unless you need them:

- **Trigger** — what starts the acquisition. Start bit (the default) waits for the line to move. Free run grabs immediately. Trigger key waits for the front-panel TRIGGER key, as described under **The TRIGGER key** above. **If the thing you asked for never happens, the capture goes ahead anyway** after a few seconds and the note row names the one that did not arrive — edge trigger unavailable; captured free-running, or front for the TRIGGER key. Those bytes are real but they are not lined up with the event you wanted.
- **Rear BNC** — lets an external trigger in on the rear connector, and FC Out sends the flow-control credit pulse out of EXT TRIG OUT. FC Out is what turns a recording into an unlimited one; see **More than one window: flow control** above.

This setting affects FRAME captures. **A recording ignores it** and starts as soon as you press Capture — for a recording, the thing that decides when the device talks is the flow-control credit, not a trigger.

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### Files on the USB key

Two different files, and they are not the same thing:

| File                 | Written                         | Holds                                                                                       |
|----------------------|---------------------------------|---------------------------------------------------------------------------------------------|
| /usb1/bytesNNN.txt   | automatically, on every capture | the running log — every capture's bytes as text and hex. <b>NewLog</b> starts a new number. |
| /usb1/serial_NNN.txt | only when you press <b>Save</b> | a full report of what is on screen: settings, all the notes, the whole dump.                |

A recording decodes into the log file, so the file holds the whole run while the panel shows only the tail of it. The note row says so when they differ.

**Recording needs a USB key** — the file is the point of it. Normal FRAME captures work without one; they just cannot log or Save.

## When something looks wrong

| What you see                                | What it means                                                                                                         |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| line is idle (no transitions)               | Nothing is crossing the threshold. Check your connection, and check the device is actually talking while you capture. |
| no clear logic levels                       | Too much noise, or the swing is too small. 1.6 V works; 60 mV is refused outright.                                    |
| Wrong rate reported                         | The traffic may be genuinely ambiguous (below). Lock the rate if you know it.                                         |
| Bytes wrong but ERR 0, with the rate locked | The device is not running at the rate you locked. The note row warns when the line disagrees by more than 4 %.        |
| A baud rate that is not a round number      | It is reporting what it measured, because the measurement was not within 2 % of a standard rate.                      |

**Some traffic is genuinely ambiguous, and no decoder can fix it.** Eight 0x00 bytes at 9600 baud 7N1 make exactly the same waveform as four 0x08 bytes at 4800 baud 8N1. The app tells you a rival rate fits rather than pretending to know. Irregular gaps between bytes remove the ambiguity completely.

## What it can cope with

Measured at 8 samples per bit. Past every one of these limits the app **reports** the problem rather than quietly getting it wrong, except where noted.

| Condition                              | Works up to                                                                 |
|----------------------------------------|-----------------------------------------------------------------------------|
| Wrong clock, rate <b>locked</b>        | $\pm 4$ %. Beyond that a fast device corrupts bytes with nothing to show it |
| Wrong clock, <b>auto-detect</b>        | $\pm 12$ % and past — it measures instead of assuming                       |
| Jitter                                 | $\pm 15$ % of a bit; past $\pm 20$ % it can fail quietly                    |
| Noise                                  | about <b>40</b> % of the logic swing — roughly 1.3 V on a 3.3 V line        |
| Slow edges, long cables, weak pull-ups | filtering up to <b>0.7 bit times</b>                                        |
| Logic swing                            | <b>1.6 V</b> up to 5 V TTL                                                  |
| DC offset                              | anything that keeps the line inside $\pm 10$ V                              |

**Rates from 300 to 250000 baud** decode byte-exact on the bench, and so do odd ones — 900, 1500, 3600, 7200 and arbitrary values like 1379, 8123 and 104857. Two exceptions are known and both are **flagged rather than silent**: long gapless payloads at 2400 baud produce one to three damaged frames in a few hundred, and one 250000-baud run matched 497 of 499 bytes with the odd byte marked. A rate within 2 % of a standard one is reported as the standard one, because real devices use standard rates.

**Tolerance follows samples per bit, not baud rate.** A slow line is not automatically the safe one — read SA/BIT.

**If you are writing the firmware at the other end**, two things help: keep your edges within  $\pm 15$  % of a bit time (that is  $\pm 15.6$   $\mu$ s at 9600 baud but only  $\pm 1.3$   $\mu$ s at 115200, so if an interrupt can delay your transmit loop, send slower), and avoid padding with 0xFF, which gives the rate detector only one narrow pulse per frame to measure. 0x00 padding is fine.

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Measured figures, the full tolerance envelope, press-by-press timings and the reasoning behind the design are in **REFERENCE.md**.